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Working Paper WP0125

Slide Deck · Pillar: AI Agents / Kolmogorov Theory

Agent Souper

A liquid agent-system substrate for telehomeostasis and meta-agents (Beta)

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What is Agent Souper?

An open, local-first simulation platform — a working Beta — for **liquid agent systems**: Kolmogorov-Theory agents in a shared 2D world, coupled by a dynamical connectome.

- Executable counterpart of the *Agent Soups* formalism; L1/L2/L3 agent taxonomy.
- Every L3 agent is an *explicit* KT decomposition.
- Three configurable worlds: Simple Demo (L1), Predator–Prey, Soccer.
- Purpose: make telehomeostasis and meta-agent emergence empirical.

The KT agent (L3)

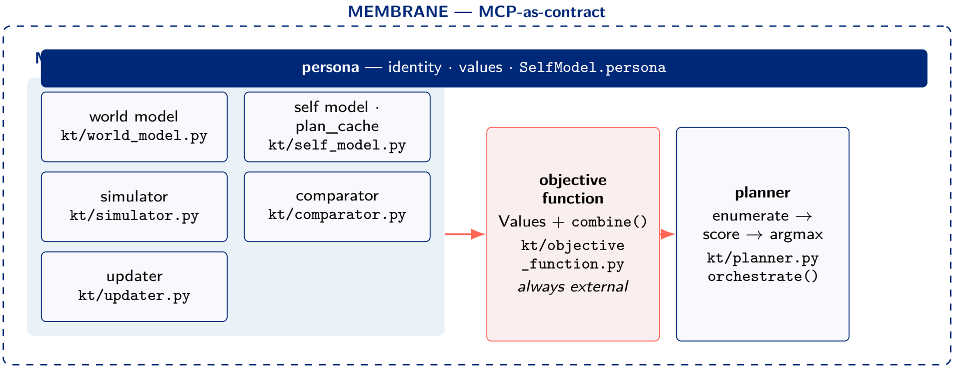
L3 = persona + 7 modules + swappable membrane

- Model Engine: world model, self model, simulator, comparator, updater.
- **Objective Function** — external, never fused into a net.
- Planning Engine: the planner.

Fast loop, every tick — enumerate, score, take the argmax:

$$a^{\star} = \arg\max_a \frac{\sum_v w_v\, s_v(a)}{\sum_v w_v}, \qquad w_v = 0.5^{\,p_v - 1}\, r_v$$

The agent model, mapped to the software



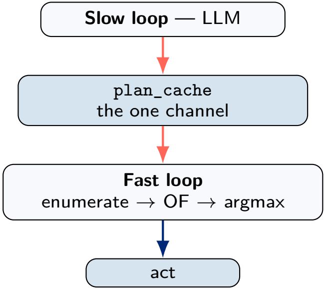
ME / OF / PE ≡ the seven `agents/kt/*.py` modules, wired by `agents/algorithmic.py`. We run the agent architecture — not an analogy.

The dual-rate contract

Fast loop — rules, ∼60 Hz, every tick, deterministic. *This is the agent.*

Slow loop — optional LLM, ∼0.5 Hz, per-agent, fire-and-forget.

The engine *never awaits* the LLM. Off ⇒ exact rules baseline, persona inert.



What the LLM actually does — one implementation option

The slow LLM writes per-Value multipliers $r_v \in [0, 5]$ into `plan_cache`. It *rescales the aggregation* in the OF combine — so the maximised scalar shifts:

$$\text{valence} = \frac{\sum_v 0.5^{\,p_v - 1} r_v \, s_v(a)}{\sum_v 0.5^{\,p_v - 1} r_v}$$

It does

- reweight Values (r_v) — changes OF *evaluation*
- self-tune its own planner params (subgoal, step)

It does not

- change the score functions $s_v(a)$, add/remove Values
- choose actions; `official_valence` is display-only

Remark: Not a KT requirement

KT only requires the OF be *external and scalar*. This LLM-as-OF- reweighter is one design choice — one could leave the OF fixed, propose new Values via the closed loop, or tune the model instead.

Telehomeostasis — without a conductor

Definition: Telehomeostasis (operational)

Each member carries an objective whose argument is the persistence of the *kind*, acted on from *local* information only — a compressed collective objective, not a directive from a controller.

- No collective orchestrator — the LLM sees only what its one agent sees, as in nature.
- Local ingredients: persona, the OF stack, and a shared observable (the scoreboard) each agent perceives.
- The score is *observation-only*; coordination must *emerge*.

What the soup shows — reported honestly

Soccer (EXP-001)
Telehomeostatic team-play beats identical selfish 7/8, 23–2 ($\sim$11$\times$) — but only on an honest substrate (the sign flipped five times).

Predator–Prey (EXP-002)
Sacrificial prey persist 0.57$\times$ selfish, 0/8 — hypothesis **refuted**. Altruism's value is contingent.

Remark: The methodological result

Do not believe a collective-objective effect until the substrate can no longer flatter it. The refutation is a *result*, not a failure.

Toward meta-agents

Hypothesis: Meta-agent emergence

A collective is a KT meta-agent when its members each internalise a shared objective, observe a shared signal, and reason locally — with *no* central orchestrator. Meta-agency is *measured*, not imposed.

- A telehomeostatic team is a candidate: shared goal + shared observable + local reasoning ⇒ emergent competence (EXP-001).
- Open programme: estimate the collective's model compression and its AMI with its own future — the agenthood threshold.

Roadmap: collective-agenthood metric · forward-model self-tuning · networked-MCP lift · richer world.

References

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